PHYS101/Fall 2019 - 2020 FINAL

- This is a closed-everything exam.
- Calculators or computing devices are neither allowed nor needed.
- Exam duration is **120 minutes**.
- Consists of a conceptual question and five problems worth of **120 points**.
- Answers without work shown or proper explanation will not be given any credit.

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

PHYSICS 101 NO: _____

NAME: _____

Questions	Problem 1	Problem 2	Problem 3	Problem 4	Problem 5	Total

Formula Sheet

Average and Instantaneous Quantities: If f is rate of change of g wrt t:

$$f_{av} = \frac{g(t_f) - g(t_i)}{t_f - t_i} \ , \quad f(t_i) = \frac{dg}{dt}\,|_{t=t_i} \ .$$

Unit Vector Notation:

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$

Scalar Product:

$$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$$

$$\vec{A}\cdot\vec{B} = |A||B|\cos\phi$$

Vector Product:

$$\vec{A}\times\vec{B} = \vec{C}$$

$$C_x = A_yB_z - A_zB_y$$

$$C_y = A_zB_x - A_xB_z$$

$$C_z = A_xB_y - A_yB_x$$

$$|\vec{A}\times\vec{B}| = |A||B|\sin\phi$$

Constant x-acceleration:

$$v_x = v_{0x} + a_xt$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_xt^2$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

Uniform Circular motion:

$$a_{\text{rad}} = \frac{v^2}{R} \quad T = \frac{2\pi R}{v}$$

Relative velocity:

$$v_{\text{P/A-}x} = v_{\text{P/B-}x} + v_{\text{B/A-}x}$$

$$\vec{v}_{\text{P/A}} = \vec{v}_{\text{P/B}} + \vec{v}_{\text{B/A}}$$

Newton's Laws:

$$\vec{R} = \vec{F}_1 + \vec{F}_2 + \ldots = \sum\vec{F}$$

$$\sum\vec{F} = 0, \ \sum\vec{F} = m\vec{a}$$

$$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}}$$

Friction Forces:

$$f_k = \mu_k n \quad f_s \le \mu_s n$$

Work done by a force:

$$W = \vec{F}\cdot\vec{s} = Fs\cos\phi$$

$$W_F = \int_{\vec{x}_1}^{\vec{x}_2}\vec{F}\cdot d\vec{x} \text{ (varying F)}$$

Work and Kinetic Energy:

$$W_{\text{tot}} = K_2 - K_1, \ K = \frac{1}{2}mv^2$$

Power:

$$P_{\text{av}} = \frac{\Delta W}{\Delta t} \ , \quad P_{\text{av}} = \vec{F}\cdot\vec{v}$$

Gravitational Potential E.:

$$\Delta U_g = -W_{\text{grav}} = mgy_f - mgy_i$$

Elastic pot. energy:

$$\Delta U_{el} = -W_{\text{el}} = \frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2$$

Force from potential energy:

$$\vec{F} = -\left(\frac{\partial U}{\partial x}\hat{\imath} + \frac{\partial U}{\partial y}\hat{\jmath} + \frac{\partial U}{\partial z}\hat{k}\right)$$

Conservation of energy:

$$K_1 + U_1 + W_{\text{other}} = K_2 + U_2$$

or: $\Delta K + \Delta U = W_{\text{other}}$

Momentum of a particle:

$$\vec{p} = m\vec{v} \ , \quad \sum\vec{F} = \frac{d\vec{p}}{dt}$$

Impulse & Impulse-Momentum:

$$\vec{J} = \sum\vec{F}\Delta t \ , \quad \vec{J} = \Delta\vec{p} = \vec{p}_2 - \vec{p}_1$$

$$\vec{J} = \int\vec{F}dt \text{ (varying F)}$$

Conservation of momentum:

$$\vec{P} = \vec{p}_A + \vec{p}_B + \ldots \ , \quad \vec{P}_i = \vec{P}_f \ (\vec{P} = \vec{P}')$$

Elastic collision:

P and K of the system are conserved.

Inelastic collision:

P is conserved, K is NOT conserved.

Elastic Collision in 1D:

$$v'_{1x} = \frac{2m_2v_{2x} + (m_1 - m_2)v_{1x}}{m_1 + m_2},$$

$$v'_{2x} = \frac{2m_1v_{1x} + (m_2 - m_1)v_{2x}}{m_1 + m_2}.$$

Center of mass:

$$\vec{r}_{\text{cm}} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \ldots}{m_1 + m_2 + \ldots}$$

Rotational Kinematics:

$$\omega_z = \frac{d\theta}{dt} \ , \quad \alpha_z = \frac{d\omega_z}{dt}$$

$$\theta = \theta_0 + \omega_{0_z}t + \frac{1}{2}\alpha_zt^2$$

$$\theta - \theta_0 = \frac{1}{2}(\omega_{0_z} + \omega_z)t$$

$$\omega_{f_z}^2 - \omega_{0_z}^2 = 2\alpha_z(\theta_f - \theta_0)$$

Relating Linear and Angular Kinematics

(Rolling without Slipping/Sliding):

$$v = r\omega \ , \quad a_{\text{tan}} = r\alpha$$

$$a_{\text{rad}} = \frac{v^2}{r} = \omega^2r$$

Moment of Inertia and Rotational Kinetic Energy:

$$I = m_1r_1^2 + m_2r_2^2 + ... = \sum_i m_ir_i^2,$$

$$K_{rot} = \frac{1}{2}I\omega^2$$

Parallel Axis Theorem:

$$I_P = I_{\text{cm}} + Md^2$$

Work Done by a Torque:

$$W_\tau = \tau_z\Delta\theta \ \text{ (const. torque)}$$

$$W_\tau = \int_{\theta_1}^{\theta_2}\tau_zd\theta \ , \quad P = \tau_z\omega_z$$

Work-Energy Theorem in Rotation:

$$W_\tau = \Delta K_{rot} = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$$

Angular Momentum:

$$\vec{L} = \vec{r}\times\vec{p} = \vec{r}\times m\vec{v} \ \text{ (for a particle)}$$

$$\vec{L} = I\vec{\omega} \ \text{ (for a rotating rigid body)}$$

Rotational Dynamics and Angular Momentum:

$$\sum\vec{\tau} = \frac{d\vec{L}}{dt}$$

$$\sum\vec{\tau} = 0 \Rightarrow \vec{L} : \text{constant } (\vec{L}_i = \vec{L}_f)$$

Questions: *(20 points)*

Explain your answers! No credit will be given for answers without explanations.

(Q1) (7 p) A non-zero net force acts on a point-like object. Are the following statements true or false?

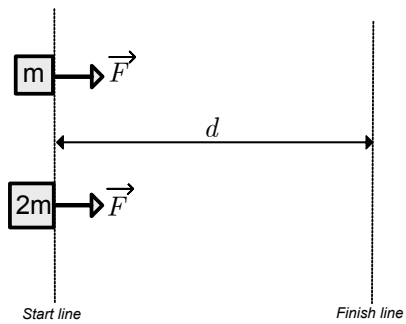
In general,

i. If $\vec{F}_{net}$ is parallel to $\vec{v}$, speed either increases or decreases. v and $\vec{v}$ changes.

ii. If $\vec{F}_{net}$ is perpendicular to $\vec{v}$, $\vec{v}$ changes its direction and v remains the same.

(i) The object's speed always increases.
According to above explanations this statement is false.
(ii) The object's kinetic energy could stay constant.
According to above explanations this statement is true.
(ii) The object's linear momentum could stay constant.
Linear momentum is a vector quantity and $\vec{p} = m\vec{v}$. According to above explanations $\vec{v}$ changes and this statement is false.

(Q2) (7 p) Two objects with mass m and $2m$ are initially at rest standing at the start line. The objects are applied the same force $\vec{F}$ as they move the same distance d on a frictionless surface.



(i) How do their kinetic energies compare as they reach the finish line?
Works done on each object are equal. By work-energy theorem they will undergo the same change in their kinetic energies. Since they both start from rest they will have the same kinetic energy at the finish line.
(ii) How do their linear momenta compare as they reach the finish line?
According to kinetic energy formula $K = p^2/2m$, linear momentum can be expressed as $p = \sqrt{K2m}$. Since both will have the same kinetic energy K at the finish line, $p_{2m}/p_m = \sqrt{2}$.

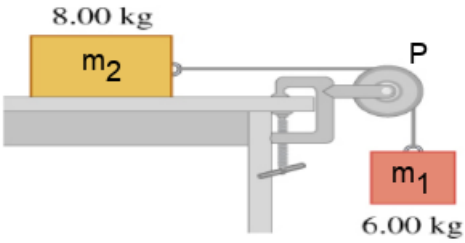
(Q3) (6 p) A solid uniform ball reaches a maximum height h_0 when it rolls up a hill without slipping. If this ball is replaced by a hollow ball with the same mass, radius and initial angular speed, will it reach a height above, below or same as h_0?
Moment of inertia of an object is proportional to mr^2. A hollow ball with the same mass has all its mass distributed near its surface. This means that the moment of inertia is larger which in turn means it is more difficult for the hollow ball to rotate, but if its rotating with same ω it has more rotational kinetic energy.

$I_{hollow,M,R} > I_{solid,M,R}$ and $v = \omega R$. Thus,

$\frac{Mv^2}{2} + \frac{I\omega^2}{2} = \frac{MR^2\omega^2}{2} + \frac{I\omega^2}{2} = Mgh$. So hollow disc with higher I will roll to a higher position,

i.e. $h_{hollow} > h_0$.

Problem 1: *(20 points)*



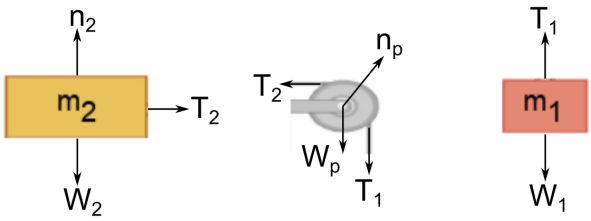
Two blocks with masses $m_1 = 6$kg and $m_2 = 8kg$, are connected by a string of negligible mass passing over a pulley of radius $r = 0.5$m. There is no friction between m_2 and the surface. (Take $g = 10$ m/s^2.)

(a) (12 p) Draw free body diagrams and determine the equations of motion for m_1, m_2, and the pulley.

For m_1: $\Sigma F_y = W_1 - T_1 = m_1 a \quad (1)$

For m_2: $\Sigma F_x = T_2 = m_2 a \quad (2)$

For pulley: $\Sigma\tau = T_1 r - T_2 r = I\alpha \quad (3)$



(b) (8 p) When the system is released from rest, m_1 moves downward with a constant acceleration of magnitude $a = 2m/s^2$, while the string does not slip over the pulley. Find the pulley's moment of inertia (I).

Insert Eq.(1) and (2) into Eq.(3), and use $a = r\alpha$.

From Eq.(1): $T_1 = m_1(g - a)$, and from Eq.(2): $T_2 = m_2 a$.

Eq.(3): $m_1(g-a)r - m_2 ar = I\dfrac{a}{r} \Longrightarrow I = \dfrac{(g-a)}{a} m_1 r^2 - m_2 r^2$.

$I = \dfrac{(10\text{m/s}^2 - 2\text{m/s}^2)}{2\text{m/s}^2}(6\text{kg})(0.5\text{m})^2 - (8\text{kg})(0.5\text{m})^2 \Longrightarrow \boxed{I = 4\text{kg.m}^2}$.

Problem 2: *(20 points)*

A 10kg satellites position with respect to a spacecraft is given as

$$\vec{r}(t) = 4\cos(\frac{\pi t}{2})\hat{\imath} - 2\sin(\frac{\pi t}{2})\hat{\jmath},$$

where origin is the position of the spacecraft and $\vec{r}$ is in meters. *(For vector quantities, give your answers in unit vector notation. Give your answers in terms of π, if required.)*

(a) (4p) Find the average acceleration vector ($\vec{a}_{av}$) of the satellite in the first 5 seconds.

$\vec{v}(t) = \dfrac{\vec{r}(t)}{dt} = -2\pi\sin(\dfrac{\pi t}{2})\hat{\imath} - \pi\cos(\dfrac{\pi t}{2})\hat{\jmath}$, then $\vec{v}_{(t=0)} = -\pi\hat{\jmath}$ and $\vec{v}_{(t=5s)} = -2\pi\hat{\imath}$

$\vec{a}_{av} = \dfrac{\Delta\vec{v}}{\Delta t} = \dfrac{(-2\pi\hat{\imath}) - (-\pi\hat{\jmath})}{5s} \Longrightarrow \boxed{\vec{a}_{av}(0-5s) = -\dfrac{2\pi}{5}\hat{\imath} + \dfrac{\pi}{5}\hat{\jmath}}$.

(b) (4p) Find the angle (θ) between position and velocity vectors at t=5 seconds.

$\vec{r}(t=5s) = -2\hat{\jmath}$ and $|\vec{r}_{(t=5s)}| = 2\text{m}$.

$\vec{v}(t=5s) = -2\pi\hat{\imath}$ and $|\vec{v}_{(t=5s)}| = 2\pi\text{m/s}$.

By scalar product: $\vec{r}\cdot\vec{v} = |\vec{r}||\vec{v}|\cos\theta \Longrightarrow (-2\hat{\jmath})\cdot(-2\pi\hat{\imath}) = 0$. Therefore $\cos\theta = 0 \Longrightarrow \boxed{\theta = \dfrac{\pi}{2}}$

(c) (4p) Find how much work (W) is done on the satellite in the first 5 seconds.

By work-energy theorem: $W_{\text{done}} = \Delta K$.

$|\vec{v}_{(t=0)}| = \pi\text{m/s}$, and $|\vec{v}_{(t=5s)}| = 2\pi\text{m/s}$

$\Delta K = \dfrac{1}{2}mv^2_{(t=5s)} - \dfrac{1}{2}mv^2_{(t=0)} = \dfrac{1}{2}(10\text{kg})((2\pi\text{m/s})^2 - (\pi\text{m/s})^2) \Longrightarrow \boxed{W_{\text{done}} = 15\pi^2\text{J}}$

(d) (4p) How much impulse ($\vec{J}$) is given to satellite in the first 5 seconds?

By impulse-momentum theorem: $\vec{J} = \Delta\vec{p}$.

$\vec{p}_{(t=0)} = m\vec{v}_{(t=0)} = (10\text{kg})(-\pi\hat{\jmath}) = -(10\pi\text{kg.m/s})\hat{\jmath}$

$\vec{p}_{(t=5)} = m\vec{v}_{(t=5)} = (10\text{kg})(-2\pi\hat{\imath}) = -(20\pi\text{kg.m/s})\hat{\imath}$

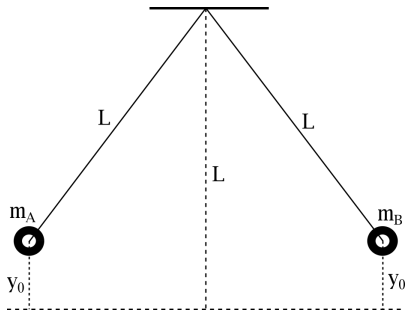
$\Delta\vec{p} = \vec{p}_{(t=5s)} - \vec{p}_{(t=0)} \Longrightarrow \boxed{\vec{J}(0-5s) = -(20\pi\text{kg.m/s})\hat{\imath} + (10\pi\text{kg.m/s})\hat{\jmath}}$

(e) (4p) Find the angular momentum vector ($\vec{L}$) of the satellite about the spacecraft at t=5s.

$\vec{L}(5s) = \vec{r}_{(t=5s)} \times \vec{p}_{(t=5s)} = (-2\hat{\jmath}) \times (-20\pi\hat{\imath}) \Longrightarrow \boxed{\vec{L}(5s) = -(40\pi\text{kg.m}^2/\text{s})\hat{k}}$

Problem 3: *(20points)*

Two objects of mass m_A and mass $m_B = 1$ kg are connected to light ropes of length L, and they are released from $y_0 = 5$ m as shown in the figure. They collide at vertical bottom. After the collision A remains at rest while B swings up to a height $y_f = 7.2$ m from the bottom.



(a) (10 p) Find the mass of pendulum A (m_A).

Apply conservation of mechanical energy for m_B after the collision.

$$\Delta K + \Delta U = 0 \Longrightarrow 0 - \frac{1}{2} m_B {v'_B}^2 + m_B g y_f - 0 = 0 \Longrightarrow v'_B = \sqrt{2gy_f} = \sqrt{2(10\text{m/s}^2)(7.2\text{m})} = 12\text{m/s}.$$

Apply conservation of mechanical energy for m_B before the collision.

$$\Delta K + \Delta U = 0 \Longrightarrow \frac{1}{2} m_B {v_B}^2 - 0 + 0 - m_B g y_0 = 0 \Longrightarrow v_B = \sqrt{2gy_0} = \sqrt{2(10\text{m/s}^2)(5\text{m})} = 10\text{m/s}.$$

Apply conservation of linear momentum for the collision. Both masses will have the same speed (10m/s) at the collision instant.

$$p_i = p_f \Longrightarrow m_A v_A - m_B v_B = m_B v'_B \Longrightarrow m_A v_A = m_B(v_B + v'_B) \Longrightarrow m_A = \frac{v_B + v'_B}{v_A} m_B$$

$$m_A = \frac{10\text{m/s} + 12\text{m/s}}{10\text{m/s}}(1\text{kg}) \Longrightarrow \boxed{m_A = 2.2\text{kg}}$$

(b) (5 p) Is the collision elastic or inelastic? Provide a valid explanation.

Check whether kinetic energy is conserved or not during the collision.

$$K_i = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 = \frac{1}{2}(2.2\text{kg})(10\text{m/s})^2 + \frac{1}{2}(1\text{kg})(10\text{m/s})^2 = 160\text{J}$$

$$K_f = \frac{1}{2} m_B {v'_B}^2 = \frac{1}{2}(1\text{kg})(12\text{m/s})^2 = 72\text{J}$$

Since kinetic energy is not conserved, the collision is inelastic.

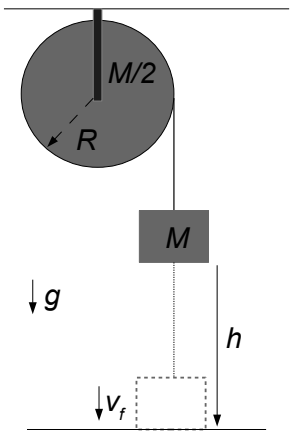
(c) (5 p) How much impulse (J) is given to pendulum B during the collision?

According to impulse-momentum theorem $\vec{J} = \Delta\vec{p} = \vec{p}_f - \vec{p}_i$.

For m_B: $J = (1\text{kg})(12\text{m/s}) - (1\text{kg})(-10\text{m/s}) \Longrightarrow \boxed{J = 22\text{kg.m/s}}$

Problem 4: *(20 points)*

A load M is attached to the end of a massless string which is wound around a disc shaped pulley with mass $M/2$ and radius R as shown. The load M is released from height of h, accelerates downward and hits the ground with a speed of $v_f = 4$ m/s. The string does not stretch and does not slip over the pulley surface.



(Take $g = 10$ m/s^2, $I_{disc} = m_{disc}R^2/2$.)

(a) (10 p) Find h in meters using the conservation of energy. (No credit will be given for solutions with any other method which does not use conservation of energy.)

For the system, Mechanical Energy, $E = K + U$ is conserved. Thus $E_0 = E_f$.

For disc, $I = (1/2)(M/2)R^2 = \dfrac{MR^2}{4}$. $E = E^{pulley} + E^M$.

Rolling without slipping condition: $v_M = R\omega_{pulley}$, i.e. $v = R\omega$.

At $t = 0$, $v_0 = 0$ and $\omega_0 = 0$. Thus $K_0^{pulley} = K_0^M = 0$.

Thus, $E_0 = U_{pulley,0} + U_{M,0} = U_{pulley,0} + Mgh$.

At $t = t_f$, $E_f = U_{pulley,f} + \dfrac{I\omega_f^2}{2} + U_{M,f} + \dfrac{Mv_f^2}{2} = U_{pulley,f} + \dfrac{MR^2\omega_f^2}{8} + 0 + \dfrac{Mv_f^2}{2}$.

Since $v_f = R\omega_f$ and $U_{pulley,0} = U_{pulley,f}$, $E_f = U_{pulley,0} + \dfrac{5Mv_f^2}{8}$

$U_{pulley,0} + Mgh = U_{pulley,0} + \dfrac{5Mv_f^2}{8} \rightarrow \boxed{\text{h}=\dfrac{5v_f^2}{8g} = 1 \text{ m.}}$

(b) (10 p) Find the magnitude of downward acceleration (a) of load M.

From kinematics, $v_f^2 - v_0^2 = 2a(h_f - h_0) \rightarrow a = \dfrac{v_f^2}{-2h} = -8m/s^2$.

Thus its magnitude is: $\boxed{a = 8m/s^2.}$

This result can also be found by drawing force and torque diagrams and solving equations of motion by using $\alpha_{pulley} = \frac{a_M}{R}$.

Problem 5: *(20 points)*

A disk of mass M and radius R is initially rotating with an angular speed of ω_0. A thin slab of mass M on a frictionless surface slowly inserted between the disk and the surface gap which fits perfectly. The disk slides on the slab for a very short time, then rolls without slipping. Finally slab looses its contact with disc with a final speed of v_f. There is no friction between the slab and the surface.

(Take $g = 10$ m/s^2, $I_{disc} = m_{disc}R^2/2$)

Find the final speed of the slab (v_f) in terms of M, R and ω_0.

For the system, the net torque is zero. Thus angular momentum of the system $L = L_{disk} + L_{slab}$ is conserved.

$L_{disc} = Iw$, and $L_{slab} = |\vec{r} \times \vec{p}| = M|\vec{r} \times \vec{v}| = MRv\sin(90^o) = MRv$.

Before the impact, $L = I\omega_0 = MR^2\omega_0/2$.

After the impact, $L' = I\omega_f + MRv_f = MR^2\omega_f/2 + MRv_f$. Using $\omega_f = v_f/R$, and $L = L'$

$MR^2\omega_0/2 = 3MRv_f/2 \rightarrow \boxed{v_f = \frac{R\omega_0}{3}.}$